

Private and social efficiency of insecticide use against an insect-borne infectious disease

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Abstract

This paper studies the economic efficiency of insecticide use against infectious pest diseases. From a model with positive spatial spillovers and negative environmental externalities of insecticides, we derive their private and social optimal uses, and propose an original flat payment as a second-best policy. We apply the model on *Flavescence dorée*, an invasive pest disease plaguing European vineyards, to show that imperfect compliance with the French policy of mandatory insecticide use provides an identification channel for the key parameters of insecticide efficiency. First, the spatial spillovers are shown to be generated according to a linear spatial econometric model. Second, the geographical indications related to wines provide an instrument for insecticide use that deals with reverse causation. Third, restricting insecticide sales to policy-approved insecticides reduces the error in measuring the quantity of insecticide used against the disease. Our results indicate that winegrowers privately underuse insecticides (with an average marginal benefit-cost ratio of about 3:1) and socially overuse them when the environmental cost is higher than 70 euro/ha/year. For a calibration of our case study, the proposed second-best flat payment is almost as efficient as the first-best spatial targeting of insecticide use.

Keywords: economics of pesticides ; vineyard management ; bioeconomic modeling ; system of equations ; spatial auto-correlation.

JEL codes: C21 ; H23 ; Q12.

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1 Introduction

Invasive species and infectious diseases are mobile public bads known for causing high economic damage (Diagne et al., 2021). They trigger epidemics, disrupt ecosystem services, and negatively affect crop production. Individual self-protection responses to outbreaks may not be sufficient to stop their spread and usually do not match socially optimal levels (Althouse et al., 2010; Liu and Sims, 2016; Costello et al., 2017). This mismatch motivates public regulation that can lead to behavioral restrictions to account for spatial spillovers. Nevertheless, as enforcement is costly and violations difficult to detect, policy compliance remains contingent upon individuals' heterogeneous incentives and perceptions (Funk et al., 2009; Bargain and Aminjonov, 2020).

In agriculture, infectious pest control is almost universally achieved through the use of pesticides, where the underprovision of protection from private decisions must be balanced against the negative externalities to the rest of society (Zilberman et al., 1991; Johnson et al., 2020). To address this issue, we develop a micro-economic model of insecticide use that first highlights the role of producers' incentives, perceptions, and spatial centrality to determine their supply of public pest control. In a second stage, by taking into account the environmental costs associated with insecticide use, the model shows that insecticides can be underused or overused depending on a limited set of parameters. We then exploit imperfect compliance with a French policy of mandatory insecticide use to identify these key parameters, from which we assess the private and social efficiency (marginal benefit-cost ratios) of actual insecticide use, and simulate counterfactual policies.

Our application is based on *Flavescence dorée* (FD), an infectious vine disease transmitted by the invasive leafhopper *Scaphoideus titanus*. The origin of the FD epidemic is now recognized as the encounter between a European bacterium and an American invasive insect (Malembic-Maher et al., 2020), which is found almost exclusively on the grapevines it feeds on. The insects are never born with the disease but carry it once they feed on a contaminated plant (Tramontini et al., 2020). Once a vine plant has been contaminated, there is an incubation period of at least one year without symptoms during which the plant is infectious (Pavan et al., 2012). Then, the main visual symptom of FD is the yellowing of the leaves, which is shared with another non-infectious disease (*bois noir*), making its surveillance more challenging and requiring laboratory analysis for confirmation. As soon as there is a contamination, FD progressively reduces the quantity and quality of grapes and almost always leads to the death of the plant within a few years (Chuche and Thiéry, 2014).

Since its first detection in France in the mid-20th century, FD has spread and is now present in 12 European countries with an estimated negative impact of 4% on the average vine yield (Tramontini et al., 2020). In France, a major outbreak led to the introduction of an ambitious control policy in 2013 (Ay and Gozlan, 2020). This new control policy establishes mandatory control perimeters (MCPs) where field surveillance for FD presence and insecticide use against its vector are mandatory. Each year, the MCPs include all vineyards in municipalities adjacent to a FD cluster identified by previous surveillance efforts.¹ Under MCPs, all winegrowers are required to use an officially approved insecticide a certain number of times per year at certain time periods. There is no official data on policy compliance, while insecticide use is currently the only way to regulate the FD epidemic (Tramontini et al., 2020). Controls for compliance are quite rare but some do take place where FD is detected, with some anecdotal evidence of regulatory penalties.²

¹Some municipalities have chosen to define MCPs at a finer scale, but these are few and data on the corresponding perimeters are not available. In the rest of the paper we assume that MCPs are defined at the municipal level.

²The maximum fine is €30,000, but after an appeal to the courts one producer had this reduced to €500, (<https://www.theguardian.com/world/2014/feb/23/french-organic-winemaker-prison-pesticide>, July 3, 2024).

Our theoretical approach is based on the combination of a spatio-temporal epidemiological model, a micro-economic model of insecticide use, and a benefit-cost analysis of policy effectiveness. The epidemiological model contains some general features of infectious diseases, which are all shared by the disease under consideration: spatial spillovers from vector mobility, an incubation period without symptoms, and an imperfect observability of presence due to limited surveillance. From a dynamic model with endogenous vector regulation by insecticides, spillover effects from neighboring infectivity, and idiosyncratic point introductions, we derive a long-run equilibrium of pest presence as a cross-sectional autocorrelation between vineyard plots from all postal codes. We then distinguish two types of winegrowers according to their static or dynamic planning horizon. These two types represent two contrasted perceptions of the spatial spillovers, which correspond to different objective functions. We compare these private strategies with that of a social planner who takes into account the negative environmental effects of insecticide use in addition to the full spatial spillovers across winegrowers. The spatial spillovers and external costs of insecticides are key to evaluate the efficiency gains of an original and theoretically consistent flat transfer (positive or negative) accounting both for spatial spillovers and environmental externalities.

Our empirical application builds on theory to build a structural spatial autoregressive model for FD detections. It accounts both for the simultaneity of insecticide use (that is both a cause and a consequence of the probability of FD detection) and the measurement errors from proxying insecticide use by insecticide sales. To address simultaneity, we argue that the spatial delineations of geographical indications (GIs, called *Appellations d'Origine Contrôlée* in French) provide plausible exogenous variations in insecticide use. Most GIs in France date back to the beginning of the 20th century (Mérel et al., 2021) and are thus expected to be free from reverse causation from the current FD presence.³ French GIs were originally designed to signal production areas of high potential wine quality (Ay, 2021). They have strong persistent effects on wine prices (Combris et al., 1997), and therefore on the economic incentives to use insecticides. To address measurement errors, we use the spatial concentration of French vineyards, the exclusive presence of the FD vector in vineyards, and the the set of approved insecticides to select restricted lists of insecticides that are more likely to be specifically used against the FD vector. By assuming that winegrowers use the recommended doses from the labels on insecticides packages, we show that including other land use areas as control variables allows to address even non-classical measurement errors. We estimate the corresponding linear spatial econometric model using instrumental variables (Kelejian and Prucha, 2007; Fingleton and Le Gallo, 2008; Kelejian and Piras, 2017).

The main theoretical contribution of our paper is the design of this flat payment as a second-best policy when plot-level targeting of insecticide use is not possible. This negative tax or positive subsidy for insecticide use is shown to deal both with positive spatial spillovers and negative environmental externalities. For static winegrowers, this payment is simple to compute because it only depends on three parameters: average spillover intensity, private and social costs of insecticides. The main empirical contribution of our paper is the estimation of the causal effect of insecticide use against an infectious disease at a national scale under real agricultural practices. This contrasts with the usual experimental analysis of field trials with small spatial coverage and artificially optimal practices (Fernandez-Cornejo et al., 1998), which are unable to take into account large-scale spatial spillovers (Carlson, 1989). The simultaneity problem between insecticide use and pest pressure is well known in agricultural economics (Lichtenberg and Zilberman, 1986; Norwood and Marra, 2003; Cobourn et al., 2011), but has yet not been addressed in the literature. Our strategy exploits the institutional peculiarity of the wine sector in France through the segmentation of vineyards

³Moreover, we include a large set of controls variables against incidental confounding variables.

by century-old wine GIs. The measurement errors we consider are typical of the usual research practice of proxying pesticide use by pesticide sales (Larsen et al., 2019), where information on the crops and pests targeted is not present in the data. These data are important to estimate large-scale spatial spillovers between producers, which cannot be based on surveys covering only part of the spatial network (Larsen et al., 2024). To our knowledge, our paper is the first to use spatial econometrics with historical sources of insecticide use.

The rest of the paper is structured as follows. The following section 2 presents the theoretical framework, section 3 the data, section 4 the empirical results, and section 5 the conclusion.

2 Theoretical framework

This section presents the epidemiological model about the spatio-temporal patterns of FD (2.1), the micro-economic model of winegrowers' optimal response in presence of a policy of mandatory insecticide use (2.2), and the efficiency framework used to evaluate pest control strategies (2.3).

2.1 The epidemiological model

The incubation of the disease means that a plant contaminated at $t - 1$ is infectious at t and shows symptoms at $t + 1$. The unobservable infectivity of a region i (among n) is a latent variable d_{it}^* that depends on the current vector population, previous contamination from neighboring infectivity, and previous point introduction. We note c_{it} the proportion of vineyards with insecticide use in i at t , which reduces infectivity by $b < 0$ through vector regulation. Contamination comes from the n_i neighbors of i , with a weight $w_{ij} = 1/n_i$ if j shares a common border with i and $w_{ij} = 0$ otherwise (including for w_{ii}). The average neighboring infectivity at $t - 1$ is $\sum_j w_{ij} d_{jt-1}^*$, which affects d_{it}^* via $r > 0$ that is the share of infected vectors crossing a boundary.⁴ The point introduction of the disease also takes place in the previous period, through the planting of infected plants or the transport of infected plant material. It is an independent zero-mean uniform variable noted δ_{it-1} .⁵

The spatio-temporal pattern of FD infectivity is:

$$d_{it}^* = bc_{it} + r \sum_j w_{ij} d_{jt-1}^* + \delta_{it-1}, \quad (1)$$

where j is summed over all regions. In addition to the effect on current infectivity through b , insecticide use c_{it} impacts the infectivity of neighbors (j) at $t + 1$, which then affects the infectivity of i at $t + 2$. With $0 < r < 1$, the strength of these feedback loops decreases with the distance in space and time between the location of insecticide use and the infectivity of i .

In period t , the infectivity level of Equation 1 determines the contamination that takes place in region i .⁶ A contamination occurs when d_{it}^* exceeds the threshold $-a$, which is detected in $t + 2$. Let d_{it} be a dummy variable indicating a detection of FD caused by a contamination in $t - 2$. For a given distribution of insecticide use (which will be endogenized in the next subsection), two

⁴A recent epidemiological study finds that the annual dispersal capacity of the FD vector results in contamination up to 300 m of the source of infection (Adrakey et al., 2023). This is sufficient for the vector to move across regions in few years, as they have an average area of 75 km² (i.e. approximately a circle of 5 km radius).

⁵In the absence of strong priors against the equality of the likelihood of point introduction between regions, the horizontal probability density function of the uniform distribution is a natural choice.

⁶The timeline differs at the plant and the region levels. For a region, infectious plants at t were contaminated at $t - 1$. They contaminate other plants at t that will be infectious in $t + 1$. Contaminated and infectious plants coexist.

sources of randomness in FD detection d_{it} remain.⁷ The first comes from point introduction, since $d_{it} = 1$ requires that $\delta_{it-1} \geq -(a + bc_{it} + r \sum_j w_{ij} d_{jt-1}^*)$. The second comes from surveillance, where only a subset of vineyards is sampled for FD detection. We consider a national sampling scheme that is representative at the region level, with freely and publicly available results for all decision makers (winegrowers and regulators).

Hence, conditional on the proportion of insecticide use on i at t , the spatial configuration of all regions j , and their past infectivity levels, the probability of detecting FD in i at $t + 2$ is:

$$\mathbb{P}(d_{it+2} = 1 \mid c_{it}, \{w_{ij}, d_{jt-1}^*\}) = a + bc_{it} + r \sum_j w_{ij} d_{jt-1}^*. \quad (2)$$

The uniform distribution of point introduction between regions yields a linear probability structure with rescaled coefficients (Amemiya, 1981) such that $-1 < b < 0$ and $0 < r < 1$ are expected in our case.⁸ The incubation period with infectivity without symptoms implies that insecticide use on i at t directly impacts FD detection on i at $t + 2$. The spatial spillovers indicate that this use of insecticide in i at t also affects the infectivity of its neighbors j at $t + 1$ (though d_{it}^*), and then (indirectly) the probability of FD detection on i at $t + 4$. There is no effect of insecticide use on i at t on the own FD detection at $t + 3$, while there is an effect in all subsequent years.

We now turn to the national scale (pooling all regions) by defining the $n \times 1$ steady-state vectors of infectivity D^* and insecticide use C (assuming for the moment they exist). By noting I the identity matrix of size n and W the $n \times n$ spatial weight matrix of elements w_{ij} , LeSage and Pace (2009) show that the expected long-run equilibrium of Equation 1 has a cross-sectional spatial autocorrelation with $\mathbb{E}(D^* \mid C, W) = b(I - rW)^{-1}C$.⁹ The infectivity of each region depends on the use of insecticide by all other regions on the long run. For a given distribution of insecticide use and a spatial configuration of regions, the vector D indicating at least one FD detection is $\mathbb{1}\{\mathbb{E}(D^* \mid C, W) + \Delta \geq -a\mathbf{1}\}$, where $\mathbf{1}$ is a $n \times 1$ column vector of 1 and Δ the random vector of point introduction. For regions (in rows), the probability of at least one FD detection is then:

$$\mathbb{P}(D = 1 \mid C, W) = a\mathbf{1} + b(I - rW)^{-1}C. \quad (3)$$

This formula is a system of n interrelated linear equations, where the sums in rows or in columns of $(I - rW)^{-1}$ are the Bonacich measures of centrality (Bonacich, 1987). The elements of the vector of insecticide use C have heterogeneous effects on the probability of detecting FD on the long run, with higher marginal effects on more central regions. Except in the case of an isolated region, a marginal increase in insecticide use in i reduces its own probability of FD presence by $b(I - rW)_{ii}^{-1} > b$ and the counterparts for all regions by $b(I - rW)_i^{-1}\mathbf{1}$, which is even higher.

2.2 The micro-economic model

For all regions within the mandatory control perimeter (MCP), risk-neutral winegrowers are required by law to use insecticide. In line with the empirical evidence presented below, we allow

⁷Because of the confusion of symptoms with *bois noir* mentioned in the introduction, visual inspection is not sufficient to detect FD in the field. The only way to “observe” FD is through field surveillance and laboratory testing.

⁸As an abuse of notation, we do not distinguish the coefficients b and r between Equation 1 and Equation 2. This does not change our analysis as latent infectivity is not scaled and only marginal probabilities from (2) are of interest.

⁹By recursive substitution of Equation 1, starting from the period $t - s$ and noting Δ the vector of δ , we have:

$$D_t^* = bC_t + \Delta_{t-1} + rW(bC_{t-1} + \Delta_{t-2}) + r^2W^2(bC_{t-2} + \Delta_{t-3}) + \dots + r^sW^sD_{t-s}^*.$$

For stationary D^* , C , and Δ , these terms can be factorized and we have $[I + rW + r^2W^2 + \dots] = (I - rW)^{-1}$ when $s \rightarrow \infty$. Taking the expectation of D^* then cancels the random point introductions (LeSage and Pace 2009, p. 26).

for imperfect compliance and we restrict regulatory control to regions with a previous detection. Winegrowers do not know whether FD is present on their own plot, they only know if the FD was detected at $t - 1$ for each region. We distinguish static and dynamic winegrowers according to their planning horizons. Static winegrowers use insecticide at t according to the uprooting risk at $t + 2$ when FD is detected.¹⁰ For a given neighboring infectivity (Equation 2), they ignore spatial spillovers and consider that the marginal effect of insecticide use is b . Dynamic winegrowers behave according to Equation 3, with a long-run horizon and a recognition of the spillover effects on their own uprooting probability. The two groups are identical in the absence of spatial spillovers ($r = 0$), which makes them very close to those of Ay and Gozlan (2020).

Although the epidemiological model is defined for regions, the micro-economic model is about an atomistic vineyard plot p of normalized size used to produce wine under constant returns to scale. In the absence of FD, the winegrower earns an annual gross margin of $m_p > 0$. Ignoring natural mortality of the vine stock, the net present value (or vineyard price v_p) of the plot is the sum of the discounted annual returns $m_p/(1 + \pi)^s$ from the next period $s = t + 1$ to infinity $s \rightarrow \infty$ (π is the discount rate). As long as $1/(1 + \pi) < 1$, the sum of discounted returns converges in the long run to $v_p = m_p/\pi$. Regardless of the planning horizon, uprooting a plant results in a loss of γ years of production, such that a FD detection on plot p at t implies an earning loss of $[1 - (1 + \pi)^{-\gamma}]v_p \equiv \ell v_p$. For analytical consideration, we define ψ_p the zero-mean iid random variable of uprooting loss deviations from the region average $v_{i(p)}$ with $\psi_p \equiv \ell(v_{i(p)} - v_p)$ for $p \in i(p)$.

A static winegrower determines the proportion c_{pt} of its plot with insecticide use according to the risk of uprooting in $t + 2$ from a contamination in t , the unitary cost of the insecticide h (product and application), and the expected regulatory cost of noncompliance f (the probability of being controlled times the amount of the fine). The objective is to minimize the expected FD loss:

$$\min_{c_{pt}} \{ \mathbb{P}(d_{pt+2} = 1 \mid c_{pt}) \ell^* v_p + h c_{pt} + (1 - c_{pt}) f d_{it-1} \}, \quad (4)$$

where $\ell^* \equiv \ell(1 + \pi)^{-2}$ because compliance in t only determines uprooting risk in $t + 2$. The winegrower is assumed to choose insecticide use according to the probability of a FD detection at the regional scale as in Equation 2, and the marginal effect of increasing c_{pt} at the plot scale is similar than the effect of increasing c_{it} at the regional scale. Hence, the first-order condition indicates that $c_{pt}^* = 1$ if $h < f d_{it-1} - b \ell^* v_p$ and $c_{pt}^* = 0$ otherwise, for a pre-determined neighborhood infectivity and FD presence in the region $i \equiv i(p)$. The winegrower fully complies when the costs of insecticides are lower than the sum of the noncompliance risk and the discounted value of the uprooting risk (recall that $b < 0$). Developing v_p and integrating over ψ_p gives:

$$c_{it}^* = \mathcal{V}[(h^* - f^* d_{it-1})/b + \ell v_i], \quad (5)$$

where $\mathcal{V}$ is the cumulative distribution function of vineyard price deviations. This yields the optimal compliance rate for static winegrowers with $h^* \equiv h(1 + \pi)^2$ and $f^* \equiv f(1 + \pi)^2$. At the region level, compliance decreases with unitary insecticide cost, and increases with regulatory cost of noncompliance, discount factor, and average vineyard price. A previous FD detection provides more incentives to comply because of regulatory controls, while compliance rate increases with insecticide efficacy b if $h > f$ for a region with a previous detection. In the reverse case with $h < f$, the fine amount implies that insecticide use decreases with their efficacies against the vector.

¹⁰Uprooting infected plants detected by surveillance is mandatory but would also be enforced absent the mandate, because leaving an infected plant in place is inefficient for winegrowers. For a small probability of recovery, it increases the contamination risk for the other vine plants of the plot.

Contrary to the previous static winegrower, a dynamic winegrower aims to minimize the net present value of the income stream, taking into account the long-run equilibrium of the uprooting probability (Equation 3). To simplify the discounting scheme, we assume that uprooting occurs at most once per plot, so that the current discounted loss of a detection is $\mathbb{P}(d_{pt} = 1 \mid c_{pt})\ell^*v_p / [\mathbb{P}(d_{pt} = 1 \mid c_{pt}) + \pi]$.¹¹ Thanks to the atomistic structure of the vineyard plots, the previous FD detection d_{it-1} at the region level i is taken as given. Dynamic winegrowers have an infinite planning horizon, with discounted streams of insecticide costs and expected regulatory fines that cancel in the first-order conditions. The use of insecticide from the minimization program also has a bang-bang structure (i.e. only corner solutions) with:

$$c_{it}^{**} \approx \mathcal{V}\{[(h^* - f^*d_{it-1})/b]/(\mathbf{I} - r\mathbf{W})_{ii}^{-1} + \ell v_i\}. \quad (6)$$

The approximation is due to the simplification of the discounting scheme, which now accounts for the dynamics of the uprooting risk.¹² The previous Equation 6 becomes exact as the probability of FD detection tends to zero at the plot level, which does not contradict a positive probability at the region level due to the atomistic structure of vineyards. Comparing static and dynamic winegrowers, it appears that the average compliance rates for dynamic ones differ by the diagonal terms of the inverse matrix of spatial spillovers. The feedback loops from insecticide use appear more clearly from the infinite sum of neighboring orders as $(\mathbf{I} - r\mathbf{W})_{ii}^{-1} = [\mathbf{I} + r\mathbf{W} + r^2\mathbf{W}^2 + \dots]_{ii}$ (LeSage and Pace, 2009). This shows that the diagonal elements of the inverse matrix are all greater than one if $r > 0$, so the marginal effects of all the parameters have the same signs than for static winegrowers, while they are higher in absolute values. The only difference for dynamic winegrowers is the role of their spatial configuration that increases insecticide use, increasingly with their spatial centrality.

2.3 Efficiency analysis

We now consider a social planner whose objective is to minimize FD losses both for winegrowers and for society by taking into account the negative environmental impacts of insecticide use. One insecticide use in a given normalized vineyard plot has a constant social cost of κ (in addition to the private cost for the winegrowers), and the revenues from noncompliance fines are redistributed so that they are only a neutral transfer and do not enter into the objective of the social planner. Even without external costs, the social marginal effects of insecticides are different from (static and dynamic) private programs that only account for the uprooting risks on the plots on which they are used. The social planner recognizes that the use of insecticide on i at t reduces the uprooting risk of the neighbors j at $t + 3$, irrespective of insecticide use of the latter. In addition to the environmental costs of insecticides, the planner minimizes the discounted value of the expected losses for all winegrowers from the non-diagonal elements of the spatial centrality matrix.

¹¹For a constant probability of uprooting ϕ and a loss of income ℓ^*v , the present value of the uprooting risk is:

$$\phi\ell^*v + \frac{1-\phi}{1+\pi}\phi\ell^*v + \left[\frac{1-\phi}{1+\pi}\right]^2\phi\ell^*v + \dots = \phi\ell^*v \sum_{s=0}^{+\infty} \left[\frac{1-\phi}{1+\pi}\right]^{t+s} = \frac{1+\pi}{\pi+\phi}\phi\ell^*v = [\phi/(\phi+\pi)](1+\pi)\ell^*v.$$

¹²Using the notation from the previous footnote, the derivative of the expected loss of income w.r.t. insecticide use is:

$$\left[\frac{1+\pi}{\pi+\phi}\phi\ell^*v\right]' = \left[\frac{\pi+\pi^2}{(\pi+\phi)^2}\right]\phi'\ell^*v \approx \phi'\ell^*v/\pi \quad \text{for } \phi = 0.$$

The linearity of the social objective at the vineyard plot level still gives a corner solution, with a vector of first-order conditions such that $C^{***} = \mathbb{1}[b(\mathbf{I} - r\mathbf{W})^{-1}\ell^*V + (h + \kappa)\mathbf{1} \leq 0]$ where V is the vector of plot-level vineyard prices. The optimal proportions of insecticide use are obtained by developing V and integrating over the deviations from region averages (in rows) with:

$$c_{it}^{***} \approx \mathcal{V}[(1 - r)(h^* + \kappa^*)/b + \ell v_i]. \quad (7)$$

The spatial configuration of regions does not appear explicitly because private and social costs of insecticides are constant and $\mathbf{W}$ sums to one by rows. Instead, spillover effects induce a mitigation factor $(1 - r)$ on insecticide costs to account for their positive protective effects. The only remaining source of heterogeneity across regions is the average vineyard price v_i , which is a stationary net present value of expected incomes that leads to steady state insecticide use.

In general, private insecticide use (static or dynamic) is not aligned with the social optimum and may be too high or too low, depending on the balance between positive spatial spillovers and negative social externalities. For static winegrowers, insecticide use is optimal when $fd_{it-1} = -\kappa + r(h + \kappa)$, which gives the optimal level of expected fine for regions with a prior detection (i.e. for $d_{it-1} = 1$). In the absence of external costs ($\kappa = 0$), the optimal fine for noncompliance is equal to r % of the private cost of one insecticide use, while in the absence of spatial spillovers ($r = 0$) the optimal fine is a flat spraying tax equals to the value of the environmental externality. These results highlight the main trade-off for the public regulation, with a subsidy for insecticide use when $\kappa < rh/(1 - r)$ and a flat tax otherwise. In the case of equality, both effects cancel each other out and insecticide regulation is unnecessary, with or without taking into account previous FD detection. Optimal decentralization with dynamic winegrowers is trickier because, even if they incorporate some positive spillovers, the optimal payments depend on region centrality and require a spatial targeting. With flat second-best (negative or positive) payments for insecticide use, a spatial mismatch between supply and demand for protection appears.

A first assessment of the average efficiency of private insecticides use can be obtained by comparing average benefits and average costs of a marginal increase in insecticide use. The three following $n \times 1$ vectors give the net discounted expected marginal benefits of a perpetual insecticide use (respectively for static winegrowers, for dynamic ones, and for the social planner):¹³

$$\begin{cases} MB^* = -b\ell^*\pi V \\ MB^{**} = -b\mathbf{I} \odot (\mathbf{I} - r\mathbf{W})^{-1}\ell^*\pi V \\ MB^{***} = -b(\mathbf{I} - r\mathbf{W})^{-1}\ell^*\pi V, \end{cases} \quad (8)$$

where $0 \leq MB_i^* \leq MB_i^{**} \leq MB_i^{***}$ because $b < 0$. The differences between the three rows rely on the parameters b and r that will be estimated econometrically. Since V and $\mathbf{W}$ are observed, the remaining unknown parameters are π and γ (through ℓ^*) for the marginal benefits, and h and κ for the marginal costs. We use plausible values for the first three, and a wide range of conceivable values for κ because the environmental costs of insecticides are not currently known. The overall ratio of the area-weighted average marginal benefits to the marginal costs gives the average return on a monetary unit invested in insecticides in the average region. A ratio greater than one indicates under-use of insecticides, while a ratio less than one indicates over-use.

Secondly, we propose to evaluate the French policy of mandatory insecticide use according to (i) its effective implementation with actual compliance rates, (ii) some counterfactual policies with the second-best flat payments described above, and (iii) the first-best policy of targeted payments

¹³We note $\odot$ the Hadamard product, also known as the element-wise product between two matrix of the same size.

accounting for the spatial configuration of winegrowers. The following system (9) gives the net benefits nb obtained by multiplying the difference between marginal benefits and marginal costs by the corresponding insecticide uses. As it can be seen from the equations (where S_{0i} is the vineyard area of i), their distribution depends on the considered values of the external costs κ . The term $nb^o(\kappa)$ corresponds to scenario (i), where $\hat{C}_i$ is the compliance rate accounting for measurement errors, $nb^*(\kappa)$ and $nb^{**}(\kappa)$ correspond to policy (ii) for static and dynamic growers respectively, and $nb^{***}(\kappa)$ corresponds to the first-best policy (iii). For i summing on regions, we have:

$$\begin{cases} nb^o(\kappa) = \sum_i [MB_i^{***} - (h + \kappa)] s_{0i} \times \hat{C}_i \\ nb^*(\kappa) = \sum_i [MB_i^{***} - (h + \kappa)] s_{0i} \times \mathbb{1}[MB_i^* > (1 - r)(h + \kappa)] \\ nb^{**}(\kappa) = \sum_i [MB_i^{***} - (h + \kappa)] s_{0i} \times \mathbb{1}[MB_i^{**} > (1 - r)(h + \kappa)] \\ nb^{***}(\kappa) = \sum_i [MB_i^{***} - (h + \kappa)] s_{0i} \times \mathbb{1}[MB_i^{***} > (h + \kappa)]. \end{cases} \quad (9)$$

All those net benefits are expressed in terms of social benefits, so the four equations have the same first parts on the right-hand side. In contrast, the second parts concern insecticide choices, which depends on the decision maker under consideration.¹⁴ The first row about actual practices includes predicted compliance rates to account for measurement errors from insecticide sales. The second and third rows integrate the (positive or negative) second-best payments derived from the theory. The fourth row is the social first best, which always presents higher net benefits because the regions with insecticide use are sorted according to their net contribution to social welfare. As κ increases, regions with a lower contribution to net benefits are increasingly required not to use insecticides, which is the optimal control strategy. Compliance with mandatory insecticide use is expected to be more efficient for low κ , while imperfect compliance could be welfare-improving for high values of the environmental externalities. For $\kappa \rightarrow \infty$, the second-best optima are expected to converge to the social first best, with an increasing spraying tax that increasingly discourages insecticide use. The comparison between $nb^*(\kappa)$ and $nb^{***}(\kappa)$ gives the value of the spatial mismatch between first- and second-best payments, which is expected to decrease with κ .

3 Data

We assemble a national data set for about 6,000 postal codes in 2016 and 2017. It covers mandatory control perimeters (MCPs), the number of annual mandatory insecticide use, and FD detection (3.1), the lists of policy-approved insecticides and associated sales (3.2), geographical indications and vineyard prices (3.3), and control variables about vineyard surface, landscape structure, winegrowers, and climate (3.4). This section concludes with some descriptive statistics (3.5).

3.1 Surveillance and policy variables

Raw epidemiological data are initially available for the 35,000 French municipalities, for each year over 2013–2017. We aggregate them at the postal code level (i.e. groups of 5.8 municipalities on average) to match the insecticide sales presented below. These data are constructed from mandatory surveillance efforts and are the official source of information for FD management. They can only be used for the years 2016 and 2017 due to a methodological change in 2015 following the 2013 policy change. The first variable of interest is the delineation of the MCPs, which we match

¹⁴To simplify, we consider counterfactual policies without conditioning compliance control on past FD detection.

with the vineyard areas to calculate postal code-specific areas under mandatory insecticide use. The second variable is the number of annual mandatory insecticide use, which varies from one to three depending on proximity to the FD cluster and local expertise. Its multiplication by the MCPs gives the mandatory area in terms of yearly insecticide use. The last variable of interest is about FD detection, equals to one if at least one tested plant was FD positive since 2013 and if, until then, at least one plant was tested positive in the annual tests (tests are not carried out every year in all municipalities).

3.2 Insecticide sales

We also use data on the physical quantities of all insecticides purchased in 2016 and 2017, located at the postal code of the buyer’s head-office. We match them with the list officially approved for use against the leafhopper category that includes the FD vector, together with the recommended maximum doses for this category (in physical units per area). These data include 108 approved insecticides while we do not know whether they are purchased specifically for use against the FD vector. Our strategy to reduce the associated measurement errors is to restrict these approved insecticides by using the spatial concentration of vineyards in France and the exclusive presence of the FD vector on vineyards. As a given insecticide can be used both against the FD vector or against other insects on other crops,¹⁵ we consider various lists of insecticides to mitigate both the risk of excluding insecticides effectively used on vineyards (type I errors) and that of including insecticides used on other crops (type II errors).

Type I errors are expected to decrease when we include more insecticides used on vineyards while type II errors are expected to decrease when we exclude insecticides with a high proportion of use for other crops. To mitigate both errors, we only include products used intensively in places specialized in vineyards, and exclude products use intensively in places without vineyards. We define specialized postal codes as those where the share of vineyards in the agricultural area is greater than 70% (excluding permanent grasslands that never receive insecticides). There are 183 such specialized postal codes with MCPs, from which we select the N most used insecticides (i.e. those for which the sales can be used to treat the largest areas). We check whether these selected insecticides are significantly purchased in postal codes where the share of vineyards in the agricultural area is below 1%. The insecticides purchased in these about 4,000 postal codes cannot be used against the FD vector, so we drop insecticides representing more than $\tau\%$ of these agricultural areas. We consider 9 lists of insecticide, for all the combination of $N \in \{10, 15, 20\}$ and $\tau \in \{1, 5, 100\}$. We put more emphasis on a short list ($N = 10, \tau = 1$) minimizing type II errors and a long list ($N = 20, \tau = 100$) minimizing type I errors in the main paper. More details on the selection of insecticide lists are given in the SM ([Appendix A](#)), jointly with a robustness analysis of our econometric results.

3.3 Vineyard price

As presented in the micro-economic model, we expect that insecticide use is driven by the potential economic losses from uprooting a contaminated vine plant, which depend on average vineyard

¹⁵The measurement errors are exacerbated when the approved insecticides are used on other crops. Insecticide use in vineyards is highly concentrated on two main targets, the FD vector and the vine moth (*Eupoecilia ambiguella*), which is on a different approved list. In addition, we remove insecticides listed against *Empoasca vitis*, another vine-related insect that does not carry FD, with associated insecticides that are not approved for use by the 2013 policy.

prices. Knowing that wine prices in France are mainly determined by about 370 geographical indications (GIs, [Combris et al., 1997](#)) and that most were designated between 1930 and 1970 ([Mérel et al., 2021](#)), we argue that the GI delineations provide plausible exogenous variations in the uprooting loss and so in the private incentives for insecticide use. This identification strategy relies on the fact that French GIs were designated well before FD becomes a serious problem and so it is impossible that they were affected by the current FD distribution. Moreover, these century-old GIs are designated at the plot level with sharp local variations ([Ay, 2021](#)) and are distinguished in the official publications of vineyard prices. Therefore, for each postal code i with vineyards, the average vineyard price v_i is calculated as (for g summing across all GIs):

$$v_i = \sum_g s_{gi} \times v_g, \quad (10)$$

where s_{gi} is the area share of the GI g in the postal code i , and v_g is the average vineyard price for the GI g in the first year of available data (i.e. 1990).¹⁶ These historical cross-sectional vineyard price variations are used as proxies for potential loss from uprooting plants following contamination, which allow to econometrically control for simultaneity between FD presence and insecticide use. The central identification restriction is that GI delineations and corresponding vineyard prices only affect FD presence through insecticide use. This is more likely to be the case for vineyard prices far in the past, while if v_g is independent of $s_{gi} \forall g$, the exogeneity of the shares s_{gi} is a sufficient restriction as for the construction of Bartik instruments ([Goldsmith-Pinkham et al., 2020](#)).

3.4 Control variables

As historical GI delineations can be incidentally correlated with omitted determinants of FD presence, we collect numerous economic and environmental variables typically used as controls in the FD literature ([Adrakey et al., 2022](#)). First, we use the 2010 French agricultural census to calculate the area of other agricultural land uses (annual crops, pastures, and other perennial crops), the number of winegrowers, their average vineyard area, their number of employees, and their average economic size. Secondly, we use the 2018 Corine Land Cover data to calculate the mean size of vineyard plots, the proportion of semi-natural areas, a landscape shape index, and the Shannon diversity index. Thirdly, we gather the grape varieties approved by GIs and use them to calculate the percentage of vineyard areas with the nine main French grape varieties (which are not equally threatened by FD, [Eveillard et al., 2016](#)). Finally, we merge historical average of climate variables at the municipal level ([Joly et al., 2010](#)) to calculate maximum temperature, cumulative precipitation, number of rainy days, solar radiation, wind speed, relative humidity, number of freezing days and evapotranspiration. To reduce the dimension of all these control variables, we run a principal component analysis (reported in [Figure A.2](#) of SI) and keep only the two principal axes in the regressions (accounting for about 50% of the variance of all controls).

3.5 Descriptive statistics

The 2013 policy change provides reliable data on FD detection at the national level (left panel A of [Figure 1](#)). From 1,287 postal codes with more than 10 ha of vineyards, 811 of them are under MCP in 2016. This represents about 550,000 ha with mandatory insecticide use (about 70% of the vineyard area). Because FD is not mandatory monitored outside MCP, the 675 postal codes that reports at least one FD detection since 2013 (and without an eradication since) are in the MCP, they

¹⁶For vineyards without GIs, we use the official vineyard prices provided at the *département* scale.

represent about two thirds of the total vineyard area. FD is widespread in the south-west (around *Bordeaux* and *Languedoc*), while some infections have also occurred in *Bourgogne*, the *Rhône* valley and the Alps. The northern vineyards of *Loire*, *Champagne* and *Alsace* are not affected.

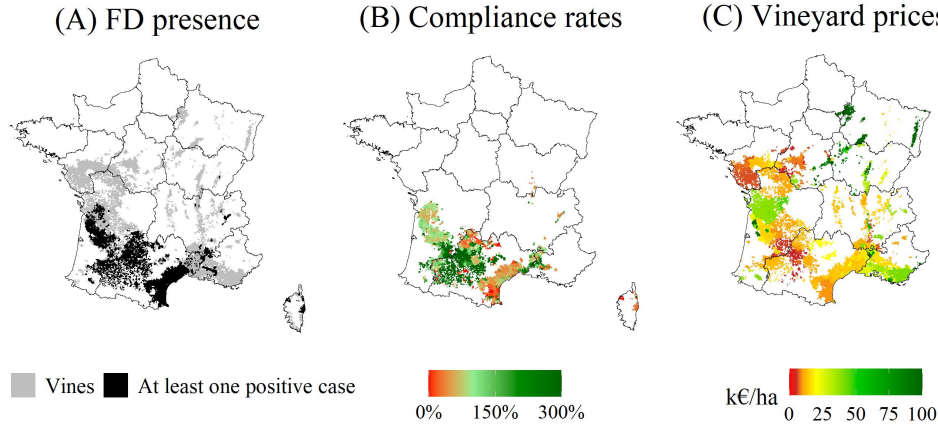


Figure 1: Distributions of (A) outcome, (B) endogenous, and (C) instrumental variables in 2016.

For the legends' clarity, compliance rates and vineyard prices are truncated at respectively 300% and €100,000/ha (although the values displayed in the maps are not truncated). Compliance rates in panel (B) are computed for postal codes within MCPs according to an intermediate list of insecticides ($N = 15$, $\tau = 5$). Panel (C) displays average vineyard prices in all postal codes with more than 1 ha of vineyards. The data for 2017 and the eight other lists of selected insecticides lead to very similar spatial patterns.

For postal codes within MCPs, we calculate compliance rates by dividing the potential treated area calculated from insecticide sales by the mandatory area with insecticide use times the number of mandatory treatments. The middle panel B of [Figure 1](#) shows the spatial distribution of compliance rates for all postal codes with a FD detection in addition to the 238 postal codes with a MCP for which FD is never detected. The right panel C of [Figure 1](#) shows the distribution of vineyard prices from the GI delineations according to [Equation 10](#). Almost a century old GI delineations produce large variations in the average vineyard price between postal codes in 1990, with an average price of €11,400/ha (in current euros) and an interquartile range of €10,000/ha. Lastly, [Table A.1](#) in SM reports the descriptive statistics of all control variables for the pool of 2016 and 2017 years. For the 811 postal codes within the MCP for at least one year, the removal of outliers (according to the double of the interquartile range) leads to a final sample of 475 postal codes (942 for the two years). This sample that is used in the regressions represents 94.6% of the vineyard area under MCP.

4 Empirical results

This section begins with the determinants of insecticide use in [\(4.1\)](#). Then, these estimations are used as first stages to instrument the effect of policy compliance on FD detection in a linear spatial econometric model [\(4.2\)](#). The economic efficiency of the mandatory treatment policy, of actual insecticide use, and of counterfactual policies are simulated in [\(4.3\)](#). A full description of the links between the theoretical framework and the empirical models is reported in [Appendix B](#) of SM.

4.1 The determinants of insecticide use

Regardless of the winegrowers' planning horizon, insecticide use is expected to depend on the vectors of FD detection D and vineyard prices V according to equations (5) and (6). The dependence on D leads to a simultaneity bias that precludes the estimation of the key efficiency parameters b and r . Furthermore, the absence of instruments in the epidemiological model precludes the identification of the structural parameters h^* and f^* on the choice of insecticides by the winegrowers.¹⁷ In turn, the linear dependence of insecticide use on the vineyard prices V , computed from a century-old GIs, provides a method to deal with the endogeneity of C and to identify the key efficiency parameters of the epidemiological model in the next subsection. This requires the additional challenge of accounting for measurement errors in the calculation of the insecticides specifically targeting the FD vector and the associated compliance rates. We propose a simple procedure that consists of including areas of other land uses as control variables.¹⁸

We consider three alternative uses of the insecticides approved against the FD vector (annual crops, temporary pastures, and perennial crops other than vineyards), with respective areas denoted S_1 , S_2 , S_3 , and with different insects targeted, so that different annual recommended doses are noted μ_1, μ_2, μ_3 . By also noting S_0 the vineyard area under MCPs, $A \equiv [S_1, S_2, S_3]$ the matrix of other land uses, and $\log(V)$ the centered vector of the natural logarithm of the vineyard price, we estimate:

$$S = \alpha_0 S_0 + \alpha_1 \overline{\log(V)} \odot S_0 + A\theta + \xi. \quad (11)$$

This specification explaining S instead of $C = S/S_0$ (in the theoretical model) is motivated by the structure of the measurement errors in the compliance rates C , which pertains to the treated area S but not to the acreage under MCPs S_0 . The measurement error on C is thus heteroskedastic and its variance is scaled by S_0 . In equation (11), we have rescaled all terms by S_0 to account for this variance structure and improve the efficiency of the estimator (see details in subsection B.1 of SM). From this specification of insecticide use we obtain the average compliance rate corrected for measurement errors directly from $\hat{\alpha}_0$. Another advantage is that we can assess the strength of the instrument V directly from a Fisher statistic on $\hat{\alpha}_1$ (because S_0 is accounted for elsewhere). The last advantage is to assess the relevance of the proposed controls for measurement errors, since each $\theta_j = \mu_j/\mu_0$ (for $j = 1, 2, 3$) is expected to be higher for land uses with more intensive insecticide use. For the available years 2016 and 2017, we pool the data to obtain a spatial panel sample of $2n$ postal codes and account for the induced correlation of errors by clustering by postal codes.

Table 1 reports the estimated coefficients for a short and a long list of selected insecticides, with and without the two principal axis of control variables described above and their spatial lags. It shows that average compliance rates range from 45% to 75% of MCP areas, depending on the list of insecticides and the covariates included. The short list of insecticides gives lower average

¹⁷The linearization of the function $\mathcal{V}$ in equations (5) and (6) does not solve this problem (see subsection B.1 of SM).

¹⁸The rationale behind this approach can be illustrated by considering an insecticide that is approved against the FD vector and has an alternative use. The observed vector of quantities purchased across postal codes is $Q = Q_0 + Q_1$, where Q_0 includes quantities used specifically on vineyards against the FD vector and Q_1 includes the quantities used on other crops against other insects. These two insecticide uses are indistinguishable from available data, while only the first Q_0 is considered in the theoretical model. Our calculation of vineyard areas with an insecticide use $S \equiv Q\mu_0^{-1}$ (where μ_0 is the recommended dose against the FD vector, taking into account the number of mandatory annual applications) also includes alternative uses not related to the FD vector. By developing Q , we have $S = C \odot S_0 + Q_1\mu_0^{-1}$ where the vector C contains the true compliance rates and S_0 are the vineyard areas within MCPs. It is clear that the measurement errors of S depend on the quantities Q_1 used in the alternative crops, which are proportional to the corresponding areas S_1 and recommended doses μ_1 if the farmers behave according to the product labels (i.e. $Q_1 = S_1\mu_1$).

compliance rates than the long list, and the average compliance rate decreases with the number of covariates included. The inclusion of covariates leads to more reasonable compliance rates than the descriptive statistics of [Table A.1](#) in SM (where the average compliance rate for the long list is 145.2%) and reduces the differences between the lists of approved insecticides. The second row of estimated coefficients shows a high importance of vineyard price in explaining insecticide use, such that a 10% increase in price increases the average compliance rate by between 0.68 and 2.9 percentage points. [Figure C.1](#) in SM displays the partial residuals associated to the instrument, which show a good linear fit between centered log vineyard price and compliance rates.

Table 1: Determinants of areas with insecticide use according to insecticide sales

	<i>Outcome: Thousand hectares with insecticide use (S)</i>					
	short list ($N = 10, \tau = 1\%$)			long list ($N = 20, \tau = 100\%$)		
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
Mandatory control area (MCA)	0.529*** (0.031)	0.490*** (0.036)	0.474*** (0.038)	0.745*** (0.046)	0.737*** (0.049)	0.732*** (0.051)
MCA $\times$ centered log vineyard price	0.158*** (0.032)	0.068** (0.031)	0.079** (0.029)	0.290*** (0.054)	0.151** (0.059)	0.154** (0.058)
Area of annual crops (000 ha)	1.234*** (0.323)	0.878** (0.310)	0.922** (0.323)	4.032*** (0.428)	3.431*** (0.390)	3.446*** (0.400)
Area of temporary pastures (000 ha)	-0.198 (0.209)	-0.227 (0.186)	-0.209 (0.182)	-0.647** (0.274)	-0.722** (0.248)	-0.717** (0.248)
Area of perennial crops (000 ha)	8.175* (5.297)	10.723** (5.526)	9.826** (5.650)	25.886*** (6.856)	27.874*** (7.664)	27.647*** (7.772)
MCA $\times$ 1st PCA axis of controls		-0.022** (0.010)	-0.056** (0.027)		-0.042*** (0.012)	-0.050* (0.035)
MCA $\times$ 2nd PCA axis of controls		0.033** (0.012)	0.032** (0.012)		0.026* (0.016)	0.025 (0.025)
MCA $\times$ spatial lag of 1st PCA axis			0.043** (0.024)			0.012 (0.029)
MCA $\times$ spatial lag of 2nd PCA axis			0.014 (0.028)			0.006 (0.037)
Number of observation	942	942	942	942	942	942
% of outcome within [0, 1]	0.711	0.711	0.711	0.435	0.435	0.435
% of prediction within [0, 1]	0.998	1.000	1.000	0.941	0.882	0.877
F stat. for weak instruments	33.530	6.563	11.864	38.738	13.046	9.383
Cragg-Donald statistic [$cv = 13.9$]	16.170	18.832	44.110	24.436	32.944	38.585

Notes: The regression sample pools 2016-2017 data for all postal codes with MCPs. Standard errors are clustered within the 475 postal codes and reported in parenthesis with *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. In column headings, N is the number of most used approved insecticides in specialized postal codes and τ is the specificity parameter used to drop insecticides used in postal codes without vineyards (see data section). The first row of coefficients gives the corrected average compliance rates, and the second row gives the marginal effect of a 1% increase in the vineyard price on compliance rates (in percentage points). The coefficients in the remaining rows are about additional conditioning variables, to account respectively for measurement errors and potential confounders. Principal components of the control variables presented [Figure A.2](#) of the SM are centered around their sample mean. The critical value cv for the Cragg-Donald statistic consider a maximum acceptable bias of 5% relative to OLS.

The coefficients associated with the other land uses indicate lower insecticide use for temporary pastures, followed by annual crops and other perennial crops, all of which are higher in absolute values for the long list of insecticides. The first two principal components of the control variables are significant at 1% in most specifications, and it appears that the first axis with high sunshine and low

altitude has a negative effect on insecticide use, while the second axis with high evapotranspiration and low humidity has a positive effect (see [Figure A.2](#) of the SM for the definition of the axes). Their inclusion in models (ii) and (v) reduces the importance of the vineyard price instrument, as evaluated by the F statistics in the bottom panel. Therefore, we consider their spatial lags in models (iii) and (vi) as additional instruments that allow to obtain a higher F statistics for the short list of insecticides.¹⁹ The critical values for the Cragg-Daniels statistics ([Andrews et al., 2019](#)) indicate that the maximum bias of weak instruments is not greater than 5% of the OLS bias in all models.

[Table C.1](#) of SM reports OLS results from an alternative specification of the determinants of insecticide use, regressing compliance rates C instead of potentially treated areas S . In addition to the efficiency argument, this alternative specification seems less appropriate for the data at hand. It produces proportions of predictions within the $[0, 1]$ interval only slightly higher than the raw compliance rates (equal to 71.1% for the short list and 43.5% for the long list), while the specifications reported in [Table 1](#) present percentages of predictions in the interval higher than 85% (see the bottom panel). Moreover, the vineyard price instrument appears to be weak in all these alternative specifications reported in [Table C.1](#) of the SM. We also interpret this result as a consequence of the mismatch of the alternative model with the data, against which we favor the [Equation 11](#) with a better goodness of fit. We predict compliance rates from $\hat{C} = \hat{\alpha}_0 + \hat{\alpha}_1 \log(V) + \mathbf{X} \hat{\theta}$ using the coefficients of [Table 1](#), where $\mathbf{A} \equiv \mathbf{X} \odot S_0$ so that $\mathbf{X}$ is the $2n \times 3$ matrix of the alternative land use areas divided by the vineyard area within MCPs, for each postal code.

4.2 The effect of insecticide use on pest presence

We now consider the estimation of the key theoretical parameters b and r for the economic efficiency of insecticides, defined by the epidemiological model and related to FD detections D in [Equation 3](#). As presented by [Boucher and Bramoullé \(2020\)](#) and discussed in more detail in [subsection B.2](#) of SM, the dummy variable D is generated by a linear probability model of compliance rates C , spatial lag of FD detection WD , and the various control variables included in the first stages. This reads as follows:

$$D = \beta_0 + bC + rWD + \mathbf{X}\eta + \varepsilon. \quad (12)$$

This type of linear probability model is increasingly used for binary outcomes in place of more common models such as logit or probit ([Angrist and Pischke, 2008](#)). It allows endogenous explanatory variables to be treated with fewer functional constraints on the reduced form of compliance rates presented in the previous subsection ([Wooldridge, 2010](#)) and on the predictions of the average FD detection in neighboring postal codes WD ([Kelejian and Prucha, 2004](#)).²⁰ We use V , WV and W^2V (and the spatial lags of the control variables in some specifications) as instruments for both compliance rates C and the spatial lag of FD detection WD , as is typically advised in the literature ([Kelejian and Piras, 2017](#)). The main debate about such linear probability models concerns the structure of the reduced errors ε , which are shown to be heteroskedastic and autocorrelated in [Equation B.2](#) of SM. We estimate [Equation 12](#) using two methods found in the literature, first through a standard 2SLS procedure with clustered standard errors ([Boucher and Bramoullé, 2020](#)) and a generalized procedure mixing 2SLS and GMM ([Kelejian and Prucha,](#)

¹⁹This result is obtained for the majority of all insecticide lists reported in the top panel of [Table C.2](#) in SM.

²⁰Because we pool data from 2016 and 2017, the square matrix W about spatial spillovers is block diagonal with size $2n$, considering that a postal code in 2016 is not a neighbor of a postal code in 2017 (and vice versa).

1998) with a heteroskedastic and autocorrelation consistent (HAC) matrix of variance-covariance (Kelejian and Prucha, 2007).²¹ The results are reported in Table 2 and Table 3 respectively.

Table 2 first reports one-stage IV estimations in columns (i) and (iv), with compliance rates directly instrumented as in the first stages of Table C.1 in SM (but with WV and W^2V as additional instruments). These instruments are strong for the spatial lag of FD detection, but remain weak for the compliance rates. Nevertheless, we obtain the expected signs for b (negative) and for r (positive), while the estimated coefficients are not statistically significant and the long list gives a high value with low economic credibility ($r > 2$). In line with the results of the previous subsection, predicting compliance rates from our favored Equation 11 gives strong instruments for columns (ii) and (v), while including control variables reduces their strength in columns (iii) and (vi), but remains significantly higher than in one-stage the IV. From the 2SLS estimations, we estimate $\hat{b}$ between -0.335 and -0.126 and $\hat{r}$ between $+0.35$ and $+0.56$. The differences depend mainly on the inclusion of control variables and their spatial lags, and the estimates are robust across insecticide lists. The middle and bottom panels of Table C.1 in SM confirms these results by showing more variation between specifications in the rows than between selected insecticide lists in the columns.

It is clear from the theory that $\hat{b}$ is the expected effect of a one percentage point increase in compliance rates on the probability of FD detection for static winegrowers. For dynamic winegrowers and the social planner, the marginal effects of insecticide use are heterogeneous across postal codes according to their spatial configurations. Following the spatial econometric literature (LeSage and Pace, 2009), the middle panel of Table 2 summarizes these heterogeneous impacts in terms of average direct impacts (adi), average indirect impacts (aii), and average total impacts (ati).²² The adi values correspond to the expected effects of a one percentage point increase in the compliance rate on the probability of FD detection for dynamic winegrowers, which are very close to the $\hat{b}$ of static winegrowers. The expected effect for the social planner ati takes into account the indirect effects aii that are not accounted for by dynamic winegrowers. The estimators are distributed between -0.285 in column (v) and -0.527 in column (ii), indicating that static and dynamic winegrowers underestimate the total impact of insecticides. Table C.2 in SM confirms these results for all lists of insecticides, where specifications (ii) and (v) give the most stable results with $\hat{b} \approx 0.2$ and $\hat{r} \approx 0.55$.

To assess the robustness of these 2SLS results, we estimate Equation 3 with an alternative generalized spatial 2SLS (GS2SLS) procedure that iteratively mixes generalized moments (GM) and IV estimations (Kelejian and Prucha, 2004; Kelejian and Piras, 2017, see Appendix B of SM for more details). As in the previous table, columns (i) and (iv) of Table 3 report S2SLS estimations from the raw instrumentation of compliance rates instead of areas with insecticide use. The instruments remain weak in the first stages, while $\hat{b}$ and $\hat{r}$ are now significant for the short list and only $\hat{r}$ is significant for the long list (with a more plausible value than that obtained previously with one-step IV). The four other columns of Table 3 report GS2SLS estimators with the predicted compliance rates from the favored first stages of Equation 11. The estimated values for the structural parameters $\hat{b}$ and $\hat{r}$ are consistent with those in the previous table, with greater robustness across econometric specifications and insecticide lists. The average impact statistics reported in the middle panel are also quite similar, with an expected effect of 10 percentage points more compliance of about -0.35 for the winegrowers (regardless of their planning horizon) and 0.4 for the social planner. Table C.2 in SM shows a higher stability of the GS2SLS results, both for b and r , in relation

²¹We retain a triangular distance-based weighting scheme with a 10 km cut-off for the HAC matrix, which is different from the row-normalized first-order contiguity of W . The HAC specification accounts for time dependence as it considers 2016 and 2017 postal code observations as neighbors (see subsection B.2 of SM).

²²The average impacts are defined such that $adi = (b/2n)\Sigma_i(I-rW)_{ii}^{-1}$, $ati = (b/2n)\Sigma_i\Sigma_j(I-rW)_{ij}^{-1}$, and $aii = ati - adi$.

Table 2: Linear probability models about the effect of compliance on FD presence

	Outcome: Dummy about FD detection (D)					
	short list (N = 10, $\tau = 1\%$)			long list (N = 20, $\tau = 100\%$)		
	(i) IV	(ii) 2SLS	(iii) 2SLS	(iv) IV	(v) 2SLS	(vi) 2SLS
$\hat{b}$	-0.620 (0.761)	-0.232** (0.086)	-0.252* (0.175)	-0.205 (0.330)	-0.126** (0.046)	-0.335*** (0.090)
$\hat{r}$	0.159 (0.639)	0.568*** (0.173)	0.352** (0.154)	2.180 (2.901)	0.568*** (0.173)	0.361** (0.123)
Share of annual crops (%)	-0.048** (0.025)	2.838** (1.055)	2.291* (1.606)	-0.003 (0.053)	5.055** (1.872)	11.509*** (3.114)
Share of temporary pastures (%)	-0.005* (0.003)	-0.462** (0.169)	-0.528* (0.365)	0.006 (0.016)	-0.817** (0.300)	-2.404*** (0.649)
Share of perennial crops (%)	0.138 (0.406)	19.246** (6.985)	25.035* (17.180)	0.006 (0.968)	32.891** (12.016)	92.830*** (25.015)
1st PCA axis of controls			-0.014** (0.008)			-0.025*** (0.007)
2nd PCA axis of controls			0.014* (0.010)			0.015** (0.005)
(Intercept)	1.181 (0.962)	0.528** (0.172)	0.718*** (0.199)	-0.742 (2.240)	0.499** (0.168)	0.836*** (0.158)
Average direct impact	-0.624	-0.261	-0.262	-0.132	-0.142	-0.349
Average indirect impact	-0.108	-0.264	-0.121	0.290	-0.143	-0.168
Average total impact	-0.732	-0.525	-0.383	0.158	-0.285	-0.517
Number of observation	942	942	942	942		
R-squared	0.076	0.091	0.076	0.099		
F-stat. (compliance rate)	33.530	11.864	38.738	9.383		
F-stat. (lag of FD detection)	34.975	19.944	34.975	27.245		

Notes: The sample pools 2016-2017 data for all postal codes with MCPs. Standard errors are clustered within the 475 postal codes and reported in parenthesis with *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. In column headings, N is the number of most used approved insecticides in specialized postal codes and τ is the specificity parameter used to drop insecticides used in postal codes without vineyards (see data section). The columns (i) and (iv) report usual one-step IV estimators from the `ivreg` function of the package `AER` in R, other columns report 2SLS estimators using the predictions from [Table 1](#) in `feols` function of the package `fixest`. The first row of coefficients gives b , the effect of increasing one percentage point of compliance rate on the probability of FD detection (in percentage points). The second row gives r the spatial autocorrelation coefficient about spillovers. The coefficients in the remaining rows are about additional control variables, to account respectively for measurement errors and potential confounders. Formulas for average direct, indirect, and total impacts are given in [footnote 22](#) in the text.

to all the considered insecticide lists.

4.3 Private and social effectiveness of insecticides

To study the effectiveness of insecticides and counterfactual policies, we favor the G2SLS estimators with controls and retain estimated values of $\hat{b} = -0.35$ and $\hat{r} = 0.2$. This implies that static and dynamic winegrowers are very similar, so we consider only static winegrowers for simplicity. Implementing the efficiency analysis presented in [subsection 2.3](#) requires the calibration of additional parameters. For the net benefits defined in [Equation 8](#), we consider a discount factor

Table 3: Spatial linear probability models about the effect of compliance on FD presence

	Outcome: Dummy about FD detection (D)					
	short list (N = 10, $\tau = 1\%$)			long list (N = 20, $\tau = 100\%$)		
	(i) S2SLS	(ii) GS2SLS	(iii) GS2SLS	(iv) S2SLS	(v) GS2SLS	(vi) GS2SLS
$\hat{b}$	-0.395*** (0.126)	-0.250*** (0.046)	-0.356** (0.156)	-0.004 (0.021)	-0.136*** (0.024)	-0.369*** (0.080)
$\hat{r}$	0.495*** (0.141)	0.206*** (0.065)	0.205** (0.072)	0.507*** (0.137)	0.185** (0.068)	0.201** (0.069)
Share of annual crops (%)	-0.035*** (0.008)	3.047*** (0.576)	3.244** (1.440)	-0.026*** (0.008)	5.453*** (0.965)	12.688*** (2.764)
Share of temporary pastures (%)	-0.004*** (0.001)	-0.497*** (0.092)	-0.745** (0.326)	-0.002** (0.001)	-0.883*** (0.155)	-2.650*** (0.575)
Share of perennial crops (%)	0.157 (0.196)	20.780*** (3.735)	35.263** (15.334)	0.352** (0.115)	35.606*** (6.105)	102.367*** (22.159)
1st PCA axis of controls			-0.016*** (0.004)			-0.026*** (0.004)
2nd PCA axis of controls			0.022** (0.008)			0.019*** (0.005)
(Intercept)	1.181 (0.962)	0.859*** (0.050)	0.898*** (0.113)	0.465*** (0.115)	0.847*** (0.051)	1.003*** (0.079)
Average direct impact	-0.430	-0.280	-0.360	-0.004	-0.152	-0.374
Average indirect impact	-0.338	-0.059	-0.088	-0.004	-0.027	-0.089
Average total impact	-0.767	-0.339	-0.339	-0.008	-0.179	-0.462
Number of observation	942	942	942	942	942	942
Variance of residuals	0.109	0.072	0.072	0.073	0.071	0.067
F-stat. (compliance rate)	4.623	33.530	11.864	4.457	38.738	9.383
F-stat. (lag of presence)	17.644	34.975	19.944	17.644	34.975	27.245

Notes: The regression sample pools 2016-2017 data for all postal codes with MCPs. Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors are reported in parenthesis with *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. In column headings, N is the number of most used approved insecticides in specialized postal codes and τ is the specificity parameter used to drop insecticides used in postal codes without vineyards (see data section). The columns (i) and (iv) report usual spatial IV estimators from the `spreg` function of the package `spdep` in R (with `model= "ivhac"`), other columns report GS2SLS estimators using the predictions from previous Table 1 in the same function with `model= "sar"`. The first row of coefficients gives b , the effect of increasing one percentage point of compliance rate on the probability of FD detection (in percentage points). The second row gives r the spatial autocorrelation coefficient about insecticide spillovers. The coefficients in the remaining rows are about additional control variables, to account respectively for measurement errors of predicted compliance rates and potential confounders. Formulas for average direct, indirect, and total impacts are given in footnote 22.

of $\pi = 4\%$ and a loss of $\gamma = 6$ years of wine production after vine uprooting to obtain $\ell = 0.210$.²³ This implies that the loss from uprooting is about 20% of the vineyard price, or about €5 845/ha on average in 2017. The average annual marginal benefit of increasing insecticide use by one ha is €81.7 for a static winegrower according to the definition of MB^* in Equation 8, and €106.6 for the social planner (i.e. MB^{**} in Equation 8). For a private cost of insecticide assumed at $h = €35/\text{ha}$ and no environmental externality ($\kappa = 0$), the average net benefit-cost ratios of insecticide use are 2.33 and 3.04, respectively, as shown in the left part of panel (a) of Figure 2.

²³We double the loss of three productions years to account for replanting costs and quality loss (Ay and Gozlan, 2020).

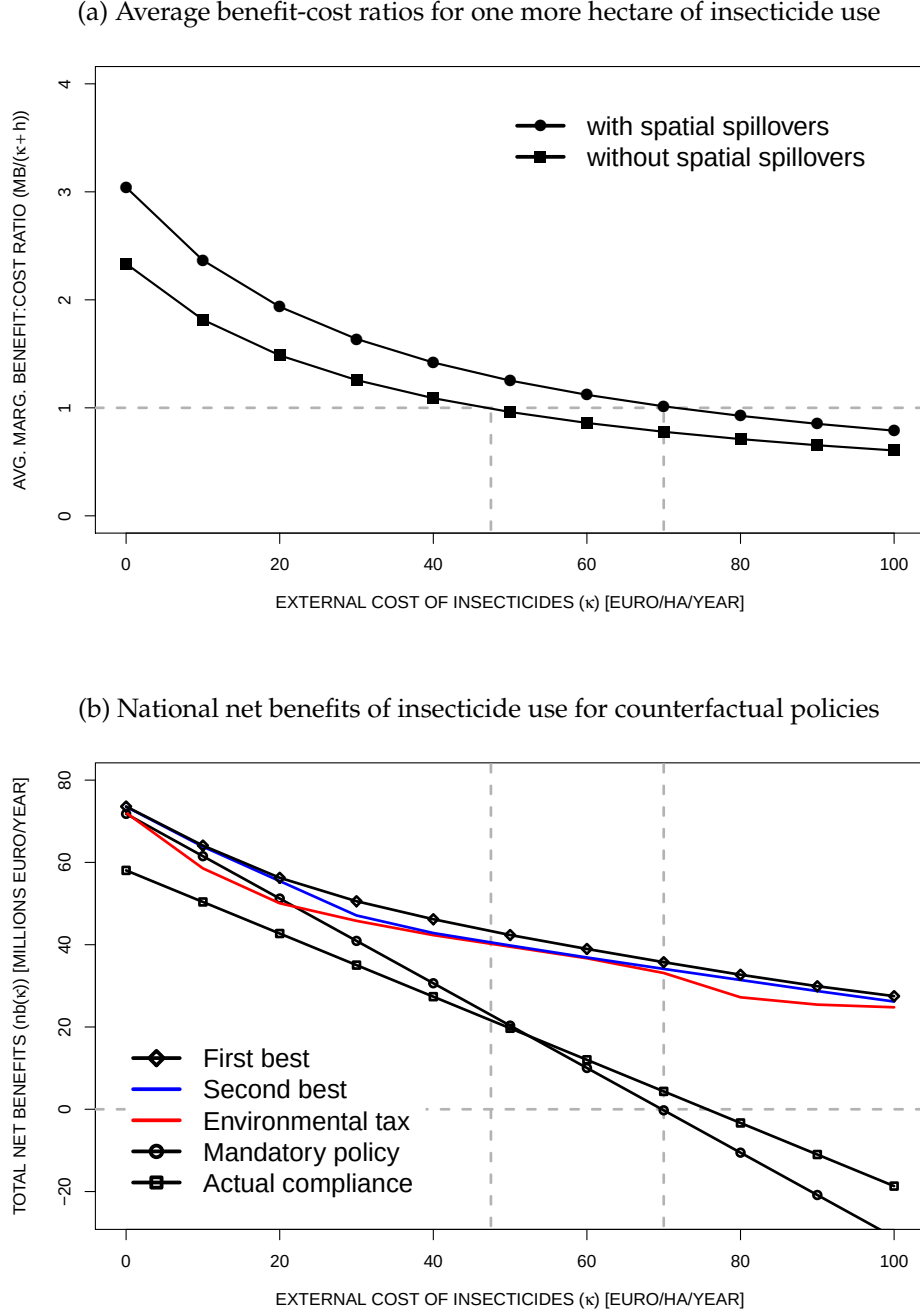


Figure 2: Social benefit-cost ratios, and net benefits of insecticide use for counterfactual policies. Parameters used: $b = -0.35$, $r = 0.2$, $\pi = 3\%$, $\gamma = 6$, and $h = 35$ for 2017 (see equations 8 and 9). Panel (a) displays the benefit-cost ratios for one ha more of insecticide use, with and without accounting for spatial spillovers. MB^{**} for dynamic winegrowers are very close to the static ones and are not reported for legend clarity. Panel (b) displays the sum of insecticide benefits less their costs at the French scale according to actual policy and compliance, social optimum, and two counterfactual policies.

Figure 2 shows how the marginal benefit-cost ratio decreases when the environmental cost of insecticide increases. For a positive environmental cost of insecticides, panel (a) shows that the marginal benefit-cost ratio decreases until it is no longer profitable to use them more. For the social

planner, it is economically efficient to increase average insecticide use as long as $\kappa < \text{€}70/\text{ha}$. For the average static winegrower who ignore the spatial spillovers, the corresponding threshold is about $\text{€}47.5/\text{ha}$. Depending on the expected regulatory cost of non-compliance with the mandatory policy f defined in the theoretical model, the average static winegrower and society would gain from using more insecticides if $f < 47.5$ and both would lose if $f > 70$. Between these two thresholds, there is a social dilemma in the use of insecticides, because it is not profitable to increase their use from a private perspective, while it is profitable from a collective perspective. Considering an environmental cost of $\kappa = \text{€}70/\text{ha}$ as an example, the average value of spatial spillovers is about $\text{€}22.5/\text{ha}$, which represents one third of the total environmental costs. Then, decentralizing the average social optimum would only require a flat tax of $\text{€}47.5/\text{ha}$ on static winegrowers (recall that insecticide doses are assumed to be constant by unit of area).

Panel (b) of [Figure 2](#) displays the net benefits of insecticide use against the FD vector at the French scale, again for different values of the environmental costs on the x-axis. The two straight lines represent respectively the imperfect (predicted from the model (vi) of [Table 1](#)) and full compliance with the policy of mandatory insecticide use. They show that the mandatory policy produces positive benefits as long as $\kappa < \text{€}70/\text{ha}$ and there is a loss of efficiency from noncompliance for $\kappa < \text{€}50/\text{ha}$. For higher values of environmental cost, the noncompliance of winegrowers increases the net benefits of insecticides, because they act as if they account for environmental externalities. Without negative externalities (i.e. $\kappa = 0$) the full compliance is close to the first best curve (computed according to the fourth row of [Equation 8](#)). However, the benefits from the policy decrease sharply when the environmental costs increase. We then simulate the benefits from our flat tax (second row of [Equation 8](#)) with respectively $r = 0$ for an environmental tax ignoring spillover, and $r = 0.2$ for the optimal tax yielding the second best equilibrium. We compare those to the benefits of the current policy. We find that our proposition is closer to the first best regardless the value of the environmental cost. The gains are significant for both low and high environmental cost (up to $\text{€}10$ millions/year) while they are lower for intermediary value (between $\text{€}30$ and $\text{€}70$ by ha).

Finding a private benefit-cost ratio strongly higher than one at the national scale may be surprising, especially because we also find a high level of noncompliance. In contrast to biological invasions causing environmental damage that are not fully perceived by winegrowers, the damages from FD are almost entirely private costs. The agents we study thus have a direct incentive to comply with mandatory insecticide applications. These findings can be reconciled by stressing (i) the importance of accounting for the heterogeneity in individual benefits from pest control, as strategic noncompliance may jeopardize policy success; and (ii) the impure public good dimension of pest control. Even if individual efforts contribute to decrease their own risks, some agents may free-ride on their neighbors' efforts. When an individual producer chooses a level of self-protection, it will not account for the positive externality it has on the neighbors (through a decreased risk from infection).

5 Conclusion

In this paper, we develop a micro-economic model of insecticide use that highlights the role of producers' incentives, perceptions, and spatial centrality to determine their supply of public pest control. Accounting for the environmental costs associated with insecticide use, the model shows that insecticides can be underused or overused depending on a limited set of parameters. We then exploit imperfect compliance with a French policy of mandatory insecticide use to identify and

estimate these key parameters, from which we assess the private and social efficiency (marginal benefit-cost ratios) of actual insecticide use, and simulate counterfactual policies.

Mandatory treatments from the pest control policy under study may seem at odds with the public health and environmental objectives to decrease pesticide use in many countries. These political objectives stand as a response to growing evidence of widespread residues and their external health and environmental costs. Although we do not estimate these external costs, our results contribute to document maximum values allowing for economically sound policies. From large-scale observational data and addressing the recurrent reverse causation bias, we show that policy noncompliance contributes to spread a infectious disease, which results in high economic losses and may further extend the duration of mandatory insecticide applications in the future. Without questioning the social benefits of curbing pesticide use, this indicates that doing so in European vineyards could be more expensive than previously thought, at least as long as alternatives for FD control do not emerge.

One key policy implication of our study is that even in favorable configurations, where regulatory obligations are designed to benefit to individuals, policy makers should not take compliance for granted. While it is clear that noncompliance can undermine the efficiency of any control policies against invasive species or infectious diseases, quantifying its significance and understanding its drivers can help the design of policy adjustments. In the case of FD control, enforcing policy requirements among many heterogeneous landowners entails a large cost of noncompliance, which supports the case for alternative regulations. Notably, a 2021 update of the French regulation on FD has granted regional authorities the power to define MCPs and allow for intra-municipal delineations. This finer scale of decision should better reflect the local benefit-cost balances. It may also exacerbate noncompliance issues because of spillovers between jurisdictions.

Finally, we argue that non-experimental approaches (like the one developed here) may help more systematically the *ex post* evaluation of pesticide use, accounting for actual farming practices and spatial interactions between neighboring farmers. Indeed, confidence in current eco-toxicological testing regimes is low because they rely on highly specific tests that are unable to encompass the full range of toxic effects that could occur with large-scale use. In the EU, environmental and health risk assessments are based on the use of a single pesticide on a specific crop, neglecting multi-product applications and many stressors that have intensified in recent years, such as climate change, habitat destruction, and increasing landscape homogeneity. Non-experimental studies on pesticide impacts, when carefully designed to establish causality, could provide evidence leading to the withdrawal of some substances with low benefit-cost ratios. More empirical research, using both experimental and observational data, is needed to design more efficient pesticide policies.

Acknowledgments

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Code and data availability

We will put all the data gathered for this research and the R code used to derive the full set of results on the INRAE dataverse once the paper is accepted for publication.

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Supplementary Material SM (not for publication)

A Data Appendix

A.1 Estimating insecticide use

Because insecticide sales are not disaggregated between targeted insects and crops, an important empirical concern is the restriction of the list of all approved insecticides against leafhoppers to those that are expected to be specifically used to comply with the FD policy. Our strategy consisted in balancing (i) the selection of approved insecticides purchased in postal codes within MCPs that are highly specialized in vineyards, and (ii) the deletion of approved insecticides purchased in postal codes without vineyards (inside and outside MCPs). This issue corresponds to a classical trade-off between a type I error from including an insecticide that is not used to comply and a type II error from excluding an used insecticide. No strategy can guarantee a zero risk for both errors, but our procedure is expected to work well because vineyards are concentrated in France because postal codes where vineyards are almost the only crop have small type I errors and postal codes without vineyards have small type II errors.

The insecticide sale data counts 108 approved insecticides over the period 2016–2017. We define the specialized postal codes as those where the vineyard share in the Utilized Agricultural Area (UAA) is larger than 70%, without including permanent grasslands. There are 183 such postal codes within MCPs, from which we select the N most used insecticides (i.e., those for which the quantities purchased can be used to treat the largest vineyard acreages).²⁴ Although we consider only the insecticides that are approved specifically against leafhoppers in vines, and then only select the insecticides that are intensively used within highly specialized postal codes within MCPs, this selection alone may not convincingly eliminate the risk of including irrelevant insecticides. Hence, we investigate whether each insecticide is significantly purchased in postal codes where the share of vines in the UAA excluding grasslands is below 0.1%. In these 4,000 postal codes (63.5%), the purchased insecticides cannot be used against the FD vector because the insect is specific to vines, so we drop among the N selected insecticides those that represent more than $\tau\%$ of UAA excluding grasslands. Our considered lists are then parameterized by N for increasing type I errors and decreasing type II errors, whereas decreasing τ do the reverse.

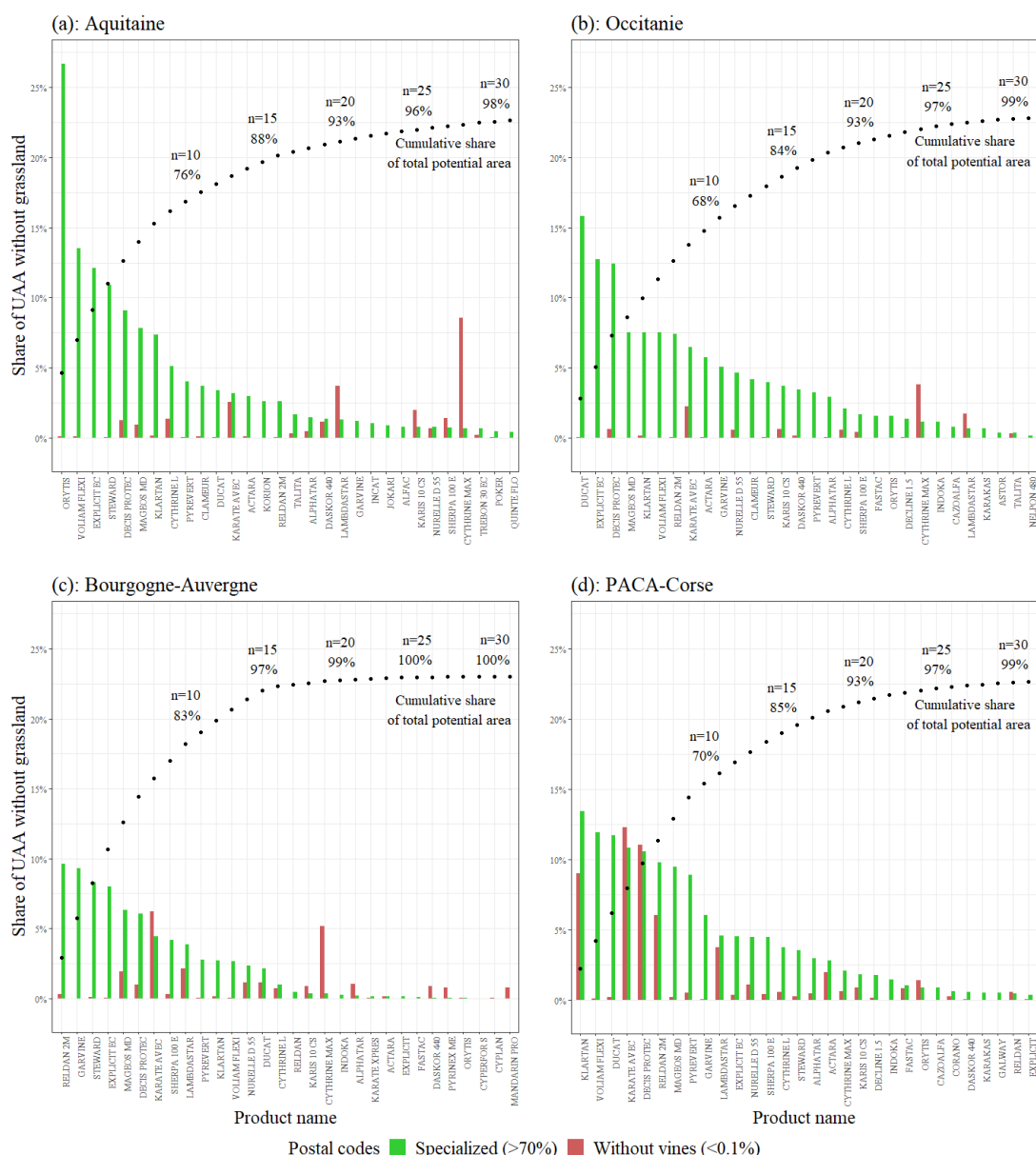
Figure A.1 shows how this strategy plays out for the four major wine regions of France. For each most-used insecticides on the x-axis, the y-axis indicates the ratio between the potential treated area and the total UAA without grassland. The green (resp. red) bar represents specialized postal codes (resp. postal codes without vines). For all regions, almost all $N = 20$ most-used insecticides are not significantly used in postal codes without vineyards, and represent a cumulative share of more than 90% of what can be treated with all the 108 approved insecticides. For the Aquitaine region of the panel (a), "CYTHRINE MAX", "LAMBDA STAR", and "KARATEAVEC" appear to be significantly purchased in postal codes without vineyards, hence they are removed for $\tau < 5$. In the econometric analysis, we evaluate the salience of this selection of insecticides by considering nine different lists with $N \in \{10, 15, 20\}$ and $\tau \in \{1, 5, 100\}$. Setting $\tau = 100$ leads to considering the top N insecticides in specialized postal codes, regardless of how they are used in postal codes without vines. This defines insecticide lists with high type I errors but low type II errors.

²⁴The conversion from the quantity purchased into actually treated areas relies on the recommended doses. For each insecticide and year, we compute the area that could be sprayed using all the quantities purchased.

Our computation of compliance rates from the different lists is the ratio between the potentially treated acreages and the acreages mandated to be treated according to the FD control policy (i.e., the product between vineyard acreages and the number of mandatory treatments within MCPs). Intuitively, setting $N = 10$ instead of $N = 15$, or $\tau = 1\%$ instead of $\tau = 100\%$, yields slightly smaller values for compliance rates. These parameters yield lists with fewer insecticides, which increases the likelihood of underestimating the true compliance rate but decreases the likelihood of overestimating it. Depending on the postal code, the bias may be positive for all compliance rates, negative for all of them, or positive for some and negative for others.

Figure A.1: Regional analysis for the restricted lists of insecticides.

Notes: The four panels represent the main wine-producing regions of France. The bars show the share of Usable Agricultural Area (UAA, without grasslands) that could be treated with the observed sales of each insecticide. We distinguish the specialized postal codes and the postal codes without vineyards, to show that insecticides with other uses can be dropped depending on τ . The dotted lines represent the cumulative shares of the most used insecticides over specialized postal codes.



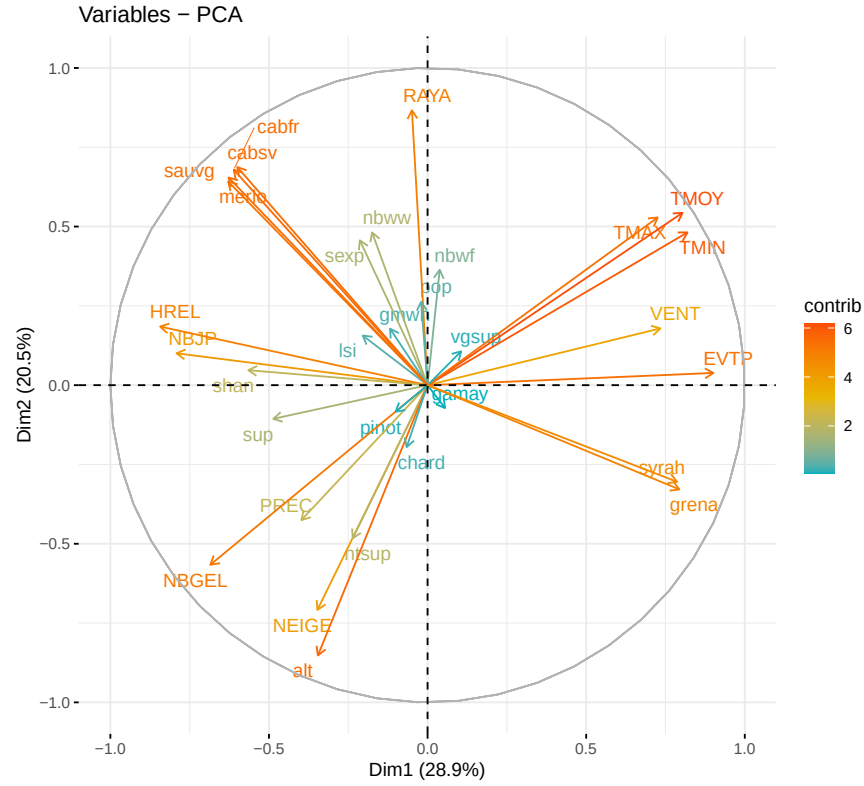
A.2 Summary statistics

Table A.1: Main variables used in the regression, pooled sample 2016-2017

Statistic	N	Mean	St. Dev.	Min	Q1	Q3	Max
FD presence	942	0.899	0.301	0	1	1	1
Log. 1990 vineyard price	942	2.798	0.646	1.609	2.431	2.931	6.108
Compliance ($N = 10, \tau = 1$)	942	0.651	0.547	0.000	0.269	0.872	3.217
Compliance ($N = 10, \tau = 5$)	942	0.810	0.679	0.000	0.361	1.049	3.281
Compliance ($N = 10, \tau = 100$)	942	0.868	0.712	0.000	0.370	1.166	3.281
Compliance ($N = 15, \tau = 1$)	942	0.774	0.654	0.000	0.334	1.021	6.277
Compliance ($N = 15, \tau = 5$)	942	0.978	0.791	0.000	0.425	1.327	3.471
Compliance ($N = 15, \tau = 100$)	942	1.047	0.826	0.000	0.447	1.442	3.471
Compliance ($N = 20, \tau = 1$)	942	1.135	4.267	0.000	0.423	1.229	126.646
Compliance ($N = 20, \tau = 5$)	942	1.379	4.316	0.000	0.486	1.575	126.646
Compliance ($N = 20, \tau = 100$)	942	1.452	4.313	0.000	0.520	1.723	126.646
Mean vineyard plot size (vgsup)	942	0.437	2.007	0.000	0.000	0.178	24.622
Percent of semi-natural area (ntsup)	942	0.313	0.261	0.000	0.090	0.493	0.986
Landscape shape index (lsi)	942	59.503	14.539	20.392	50.328	68.905	120.973
Shannon diversity index (shan)	942	2.683	1.093	0.000	1.904	3.529	4.731
Number of wine farms (nbwfr)	942	58.443	81.452	0	0	85	456
Acreage of wine farms (sexp)	942	884.615	1,246.527	0	0	1,166	6,861
Number of wine workers (nbww)	942	107.434	165.670	0	0	145	1,230
Gross margin of wine farms (gmwn)	942	600.321	954.909	0	0	894	5,649
Average temperature (TMOY)	942	13.538	1.552	5.207	12.735	14.669	16.120
Cumulative precipitations (PREC)	942	62.222	11.415	41.044	56.165	65.670	108.493
Number of rainy days (NBJP)	942	12.739	2.630	8.417	10.121	14.577	20.507
Solar radiation (RAYA)	942	83.609	2.869	73.293	81.551	86.140	88.293
Average wind intensity (VENT)	942	2.895	0.887	0.948	2.191	3.542	6.096
Relative humidity (HREL)	942	72.240	4.774	64.562	67.259	76.651	80.130
Number of frozen days (NBGEL)	942	2.702	1.345	0.028	1.750	3.304	10.844
Evapotranspiration (EVTP)	942	86.115	13.699	53.097	75.039	97.884	132.106
Postal code area (sup)	942	10.709	8.333	0.265	3.597	16.387	41.694
Thousand of inhabitants (pop)	942	8.275	8.046	0.400	3.200	10.200	71.000
Elevation (alt)	942	191.582	209.027	1	48.6	260.1	1,712
% of AOCs with Pinot N (pinot)	942	0.006	0.066	0	0	0	1
% of AOCs with Cab. franc (cabfr)	942	0.224	0.357	0	0	0.5	1
% of AOCs with Cab. sauv. (cabsv)	942	0.210	0.358	0	0	0.1	1
% of AOCs with Chardonnay (chard)	942	0.016	0.058	0	0	0.03	1
% of AOCs with Gamay (gamay)	942	0.003	0.014	0	0	0	0
% of AOCs with Merlot (merlo)	942	0.257	0.371	0	0	0.7	1
% of AOCs with Sauvignon (sauvg)	942	0.269	0.417	0	0	0.8	1
% of AOCs with Grenache (grena)	942	0.483	0.403	0	0	1	1
% of AOCs with Syrah (syrah)	942	0.479	0.404	0	0	1	1

A.3 Control variables

Figure A.2: Projection of control variables on the two main PCA axis.



Notes: The reduction of the dimension of the 31 variables along the two main principal components reflects statistical dependencies that are well known in viticulture. The first axis describes a grape ripening axis with high values for high sunshine (RAYA) and low altitude (alt). The second axis relates to moisture concentration, with high values for evapotranspiration (EVTP) and wind (VENT) and low values for relative humidity (HREL) and number of rainy days (NBJP). Temperature variables (TMOY, TMIN, TMAX) are positively correlated with both axes, as are grape varieties (Syrah, grena, cabfr, cabsv, sauv, merlo), but less so with socio-economic characteristics. [Table A.1](#) reports the full labels of the control variables reported in the figure.

B Taking the model to the data

B.1 Reduced form of compliance rates

Our first stages rely on a linear projection the compliance rates on D and V , which captures the associations suggested by the theoretical model. We write the $2n \times 1$ vector of compliance rates C^* of Equation 5 according to the following reduced form:

$$C^* = \alpha_0^* \mathbf{1} + \alpha_1^* D + \alpha_2^* V + e_1. \quad (\text{B.1})$$

The three coefficients α_0^* , α_1^* , and α_2^* are unknown, as is the $2n \times 1$ vector of errors e_1 . This reduced form describes the behavioral response of winegrowers to controls about mandatory insecticide use and to the economic incentives to comply, from which we expect $\alpha_1^*, \alpha_2^* > 0$.

The linearization of the cumulative function of vineyard price deviations implies that the regression coefficients are not directly related to the corresponding structural parameters h , f , or ℓ^* . This is not a key issue because the absence of instruments in the epidemiological model (i.e. exogenous variations of D) precludes the identification of both the structural and the approximate coefficients about the behavioral response of winegrowers. The main insights from this equation are the reverse causality from FD detections D to compliance rates C , and the economic rationale of using vineyard price V as an instrumental variable. Our empirical objective is to use Equation B.1 as a first stage in the estimation of the structural parameter b and r of the epidemiological model. We have $\mathbb{E}(V e_1) = 0$ by definition of a reduced form (Wooldridge, 2010).

The equilibrium compliance rate C^* is the ratio between the treated acreage S^* and the acreage to treat S_0 . We do not observe S^* but only proxy it by the area potentially treated by the purchased quantity of insecticide, noted S . Therefore, we do not observe the equilibrium compliance rate C^* but only a proxy $C = S/S_0$. We control the potential correlation between the measurement error in S and the instrument by including alternative land uses as covariate. Because the measurement error only affects the numerator S of C , we expect the variance of the measurement error in C to be scaled by the acreage to treat S_0 . We account for that variance structure to improve the efficiency of the estimation. We rescale equation (B.1) by S_0 to estimate equation (11) by OLS, which yields the efficient Generalized Least Square estimator of equation (B.1) (with covariates) if the variance of e_1 is proportional to S_0 .

B.2 Structural epidemiological equation

The part of the theoretical framework that we structurally estimate concerns the long-run effects of policy compliance (i.e. insecticide use) on the FD detection between postal codes. Without loss of generality, the steady state of Equation 3 can be written as $D = \mathbb{P}(D = 1 \mid C, \mathbf{W}) + e_2$ with independent errors (Boucher and Bramoullé, 2020). The additive reduced-form errors e_2 generally have no structural interpretation, while the structural parameters b and r appear explicitly in the resulting specification of FD detection (i.e. by substitution):

$$D = a(1 - r)\mathbf{1} + bC + r\mathbf{W}D + (\mathbf{I} - r\mathbf{W})e_2. \quad (\text{B.2})$$

The coefficient associated to the intercept vector is a scalar because the matrix of spatial spillovers is row-normalized according to the theoretical model. In principle, the infectivity threshold a can be identified from the estimated intercept by using the estimated value of r , but it is not of central

interest here because this threshold does not play a role in the efficiency of insecticide use under consideration. The errors of Equation B.2 are spatially autocorrelated even if $\mathbb{E}(e_2 | C, W) = 0$. This is taken into account by using both clustered standard errors as suggested by (Boucher and Bramoullé, 2020) and heteroskedastic and autocorrelation consistent matrix (HAC) of variance-covariance from Kelejian and Prucha (2007). Because we match 2016 and 2017 data, the square matrix W is block diagonal of size $2n$, considering that a postal code in 2016 is not a neighbor of a postal code in 2017 (and inversely). More importantly, the coefficients b and r , respectively such that $-1 < b < 0$ and $0 < r < 1$ according to the theoretical model, cannot be identified from ordinary least squares on Equation B.2 because of the reverse causation from Equation B.1 and the correlation of WD with neighbors' errors.

B.3 Identifying assumption

Our empirical strategy to deal with reverse causation is to use average vineyard price computed from geographical indications (GIs) as an instrument. This instrument is expected to be relevant from the theoretical model, where private insecticide use is driven by the potential loss from a FD contamination through the uprooting risk. The other requirement for its validity is the exogeneity with regard to the errors in FD presence, as specified in the following Assumption 1:

$$\textbf{Assumption 1 (exogenous instrument): } \mathbb{E}(Ve_2) = 0. \quad (\text{B.3})$$

This assumption means that vineyard prices are only correlated with FD through insecticide use.²⁵ This is likely to be true because GIs were historically designated well before FD was discovered and so cannot be caused by its current spatial distribution. Our computation of vineyard prices (Equation 10) only considers price variations between GIs, such that the variations of the instrument between postal codes are mainly driven by GI delineations. If v_g is independent to $s_{gi} \forall g$, the exogeneity of the shares s_{gi} is a sufficient restriction for Assumption 1 (Goldsmith-Pinkham et al., 2020). We also use the two principal components of a large set of control variables about agricultural and landscape structures, grape varieties, and past climate presented in Figure A.2 of SM. The projections of postal codes along these axis are expected to proxy the confounding variables correlated both with vineyard price and FD presence.

B.4 Spatial econometric methods

The other main empirical challenge for the estimation of Equation B.2 is the presence of WD among the regressors, which leads to bias in the raw IV estimation (Kelejian and Piras, 2017). Betz et al. (2020) present the asymptotic bias from using IV without accounting for spatial autocorrelation, showing that the bias is present even if V is randomly assigned, and increases with the similarity of the spatial patterns of the instrument V and the outcome D . The recognized solution from the spatial econometrics literature is to instrument the autoregressive terms by the spatially lagged values of the instrument and the control variables (Kelejian and Piras, 2017).

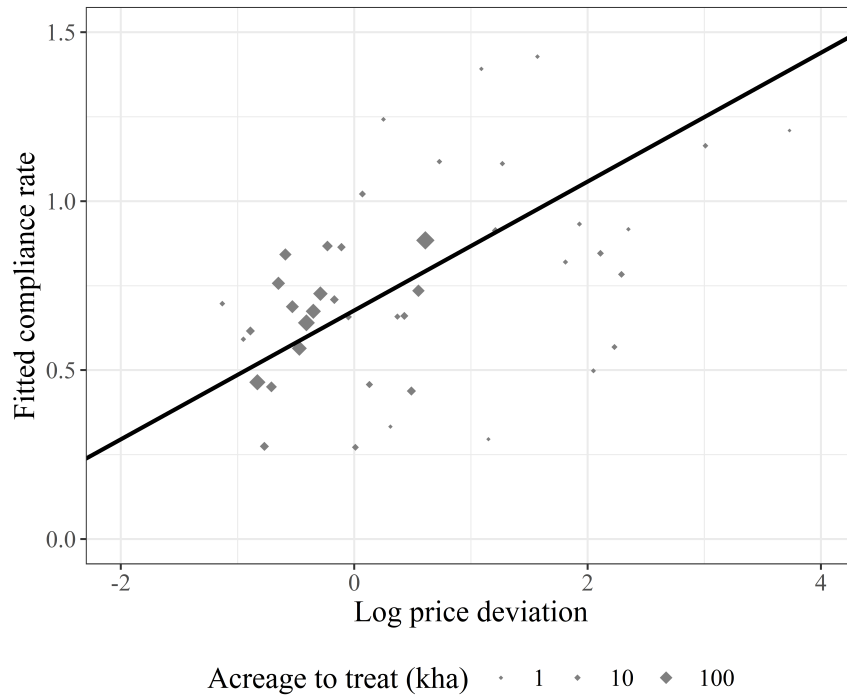
²⁵In the theoretical model, this assumption 1 corresponds to the independence of point introductions.

C Additional figures and tables

C.1 First stage predictions

Figure C.1: Partial residual plot for the instrument in the first stage.

Notes: The partial residuals come from the regression of compliance rates (with $N = 15$ and $\tau = 5$) on vineyard prices and controls for alternative land uses (rows 5–10 of Table 1 in the main text). The slope of the line is the coefficient λ_1 related to the interaction between vineyard area with MCPs and the logarithm of vineyard price deviation in [Equation 11](#). The averages are weighted by the surface to treat. The points are located in 100 regular bins spanning the interval $[-2;4]$ on the x-axis, with 53 bins containing between 1 and 144 observations. The y-value is above 1.5 for 6 observations and below 0 for 5 observations, so do not appear in the Figure. These points only account for 0.79% of the acreage to treat.



C.2 Alternative first stages

Table C.1: Determinants of compliance rates computed from insecticide sales

	Outcome: Compliance rates (% of MCPs)					
	short list ($N = 10$, $\tau = 1\%$)			long list ($N = 20$, $\tau = 100\%$)		
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
(Intercept)	0.664*** (0.039)	0.662*** (0.031)	0.663*** (0.032)	1.405*** (0.162)	1.427*** (0.151)	1.463*** (0.175)
Centered log vineyard price	0.064 (0.051)	0.007 (0.039)	0.005 (0.040)	0.118 (0.133)	0.010 (0.109)	0.005 (0.113)
Share of annual crops (%)	-0.021* (0.013)	-0.017* (0.013)	-0.018* (0.013)	-0.080 (0.235)	-0.115 (0.248)	-0.127 (0.251)
Share of temporary pastures (%)	-0.004*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	0.035 (0.054)	0.030 (0.050)	0.030 (0.050)
Share of perennial crops (%)	-0.400 (0.462)	-0.543 (0.527)	-0.538 (0.509)	0.752 (1.702)	-0.653 (2.268)	-1.136 (2.685)
1st PCA axis of controls		-0.027*** (0.008)	-0.009 (0.028)		-0.126** (0.056)	-0.008 (0.052)
2nd PCA axis of controls		0.041*** (0.009)	0.036* (0.023)		-0.065 (0.094)	0.063 (0.078)
Spatial lag of 1st PCA axis			-0.020 (0.031)			-0.143** (0.076)
Spatial lag of 2nd PCA axis			0.005 (0.024)			-0.171 (0.183)
Observations	942	942	942	942	942	942
% of outcome within [0, 1]	0.711	0.711	0.711	0.435	0.435	0.435
% of prediction within [0, 1]	0.735	0.753	0.748	0.443	0.453	0.452
F stat. for weak instruments	3.822	0.055	0.045	1.877	0.015	2.472
Cragg-Donald statistic [$cv = 13.9$]	3.605	0.348	0.342	3.474	4.023	2.900

Notes: The regression sample pools 2016-2017 data for all postal codes with MCPs. Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors are reported in parenthesis with *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. In column headings, N is the number of most used approved insecticides in specialized postal codes and τ is the specificity parameter used to drop insecticides used in postal codes without vineyards (see in the text).

C.3 Robustness checks

Table C.2: Sensitivity of main coefficients according to approved lists of insecticides

	<i>lists of approved insecticides</i>								
	<i>N = 10</i>			<i>N = 15</i>			<i>N = 20</i>		
	$\tau = 1\%$	$\tau = 5\%$	$\tau = 100\%$	$\tau = 1\%$	$\tau = 5\%$	$\tau = 100\%$	$\tau = 1\%$	$\tau = 5\%$	$\tau = 100\%$
Coefficients, clustered standard errors, and F stat. associated to the vineyard price instrument ($\hat{\alpha}_1$)									
Table 1 (i), (iv)	0.158*** (0.032) [33.530]	0.123*** (0.036) [14.339]	0.152*** (0.035) [23.953]	0.173*** (0.042) [26.884]	0.188*** (0.048) [24.885]	0.220*** (0.039) [38.865]	0.157*** (0.051) [16.639]	0.250*** (0.069) [24.540]	0.290*** (0.054) [38.738]
Table 1 (ii), (v)	0.068** (0.031) [6.563]	0.022 (0.043) [0.466]	0.066 (0.060) [3.225]	0.075** (0.032) [6.415]	0.061* (0.043) [3.308]	0.107** (0.056) [8.713]	0.046 (0.041) [2.025]	0.091** (0.045) [5.380]	0.151** (0.059) [13.046]
Table 1 (iii), (vi)	0.079** (0.029) [11.864]	0.022 (0.042) [2.539]	0.064 (0.059) [2.379]	0.083** (0.030) [13.280]	0.061* (0.042) [5.191]	0.105** (0.056) [4.992]	0.061* (0.039) [8.518]	0.097** (0.043) [10.067]	0.154** (0.058) [9.383]
Coefficients and clustered standard errors associated to to instrumented compliance rates ($\hat{b}$)									
Table 2 (i), (iv)	-0.620 (0.761)	10.602 (195.722)	-0.859 (1.387)	-0.664 (0.852)	-3.136 (22.059)	-0.523 (0.664)	-0.451 (1.788)	-0.322 (0.858)	-0.205 (0.330)
Table 2 (ii), (v)	-0.232** (0.086)	-0.297** (0.109)	-0.240** (0.088)	-0.212** (0.078)	-0.194** (0.072)	-0.167** (0.061)	-0.233** (0.086)	-0.146** (0.054)	-0.126** (0.046)
Table 2 (iii), (vi)	-0.252* (0.175)	0.012 (0.298)	-0.706*** (0.199)	-0.283** (0.165)	-0.395** (0.207)	-0.516*** (0.126)	-0.049 (0.201)	-0.268** (0.139)	-0.335*** (0.090)
Table 3 (i), (iv)	-0.395*** (0.126)	-0.137 (0.181)	-0.300** (0.137)	-0.377** (0.134)	-0.294* (0.207)	-0.313** (0.120)	0.003 (0.019)	0.004 (0.019)	-0.004 (0.021)
Table 3 (ii), (v)	-0.250*** (0.046)	-0.318*** (0.062)	-0.258*** (0.049)	-0.228*** (0.043)	-0.209*** (0.039)	-0.179*** (0.033)	-0.250*** (0.048)	-0.158*** (0.029)	-0.136*** (0.024)
Table 3 (iii), (vi)	-0.356** (0.156)	-0.186 (0.230)	-0.775*** (0.187)	-0.365** (0.150)	-0.477** (0.193)	-0.555*** (0.112)	-0.217* (0.168)	-0.331** (0.129)	-0.369*** (0.080)
Coefficients and HAC standard errors associated to the spatial lag of FD detection ($\hat{\rho}$)									
Table 2 (i), (iv)	0.159 (0.639)	8.088 (140.181)	-0.380 (1.689)	0.136 (0.725)	-2.600 (21.676)	-0.280 (1.208)	4.820 (16.442)	3.307 (7.177)	2.180 (2.901)
Table 2 (ii), (v)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)	0.568*** (0.173)
Table 2 (iii), (vi)	0.352** (0.154)	0.468** (0.162)	0.340** (0.131)	0.338** (0.153)	0.337** (0.148)	0.375*** (0.119)	0.440** (0.158)	0.331** (0.151)	0.361** (0.123)
Table 3 (i), (iv)	0.495*** (0.141)	0.456*** (0.137)	0.356** (0.140)	0.501*** (0.140)	0.388** (0.146)	0.316** (0.141)	0.503*** (0.144)	0.505*** (0.137)	0.507*** (0.137)
Table 3 (ii), (v)	0.206*** (0.065)	0.223*** (0.064)	0.210*** (0.065)	0.211*** (0.065)	0.210*** (0.065)	0.199** (0.066)	0.221*** (0.064)	0.196** (0.066)	0.185** (0.068)
Table 3 (iii), (vi)	0.205** (0.072)	0.255*** (0.070)	0.222*** (0.066)	0.211** (0.071)	0.225*** (0.069)	0.220*** (0.067)	0.236*** (0.073)	0.212** (0.070)	0.201** (0.069)

Notes: For clarity, Tables 2 and 3 in the main text only report the regression coefficients for ($N = 10, \tau = 1$) and ($N = 20, \tau = 100$). This table displays the main coefficients for all insecticide lists where the corresponding models are reported in rows. Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors of the coefficients are reported in parenthesis and Fisher statistics for the weakness of the instrument are reported in brackets (** $p < 0.001$; * $p < 0.01$; $p < 0.05$). As it is standard in the spatial econometric literature, the bottom panel does not report F statistics for the instrumentation of the spatial lag of FD presence. Spatially lagged values of the instrument and covariates are (by construction) strong predictors of the lagged values of the outcome, with almost all HAC F statistics higher than 50.